CS-423: Computer Graphics

Objective and Aim:

Computer Graphics is one of the most exciting and rapidly growing fields of Computer Science. It is basically the study of methods for generation and animation of realistic looking images and videos. It is commonly used in movies, games, television commercials, data visualizations, user interfaces and many other applications, this course is intended to introduce the fundamental principles Computer Graphics. The basic graphics pipeline is covered in this course, along with an introduction to OpenGL. This course will be conducted with an application perspective. Therefore, students should learn and implement several techniques of graphics.

Topics covered:

1. **A survey of computer graphics:-** Graphs and charts, Computer-Aided Design, Virtual Reality, Data Visualization, Education and training, Computer Art, Movies, Games and Graphical User Interfaces
2. **An overview of Computer Graphics:-** Applications, API's, Graphics Pipeline, Examples
3. **Mathematical Concepts:-** Vector Algebra, Coordinate Systems, Trigonometry and Geometry, Matrix Arithmetic, Implicit and Parametric Curves and Surfaces, Triangles
4. **Output Primitives and Attributes of Output Primitives**
5. **Raster Algorithms:-** Line Drawing, Midpoint Algorithm, Symmetry Considerations, Floodfill and Boundary Fill
6. **Geometric Transformations:-** 2D and 3D transformations, Translation, Rotation, Scaling Homogeneous Coordinates, Concatenation of Transformations
7. **Viewing:-** Projections, Orthographic Projection, Perspective Projection
8. **Illumination Models**
9. **Visible Surface Detection:-** Hidden surface removal, z-buffer, BSP Tree
10. **Lighting:-** Point Light Source, Ambient, Diffuse (Lambertian) and Specular Shading Models, Phong Shading, OpenGL lighting model
11. **Texture Mapping:-** 2D Texture Mapping, Examples, Aliasing issues, Interpolation, Bump mapping, Displacement mapping
12. **Computer Animation**

Assesement:
Assignments + Quizzes : 25
Midterm : 25
Terminal Exam : 50%
Recommended books:

- P. Shirley, S. Marschner, Fundamentals of Computer Graphics, 5/e, A K Peters, 2021
- D. Hearn, and M. P. Baker, Computer Graphics with OpenGL, 4/e, Prentice Hall, 2014
- J. D. Foley, A. van Dam, S. K. Feiner and J. F. Hughes, Computer Graphics: Principle and Practice, 3/e, Addison Wesley, 2013
- V. Scott Gorden and John Clevenger, Computer Graphics Programming in OpenGL with C++, 3/e, Mercury Learning, 2024